

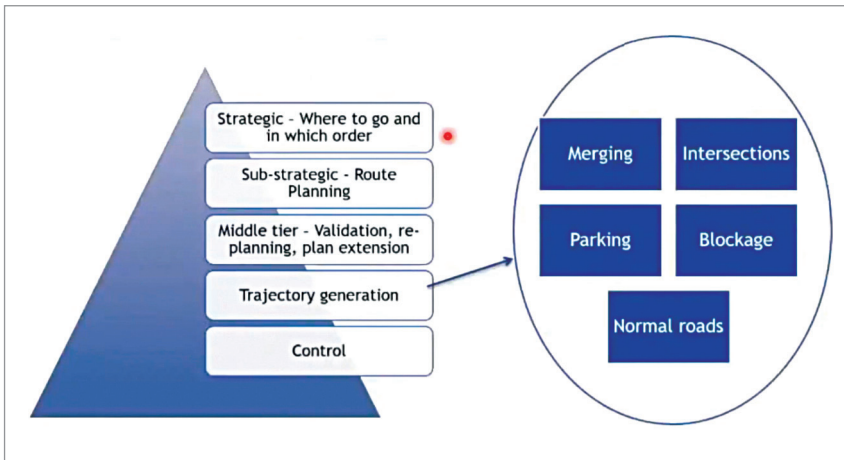


Fig. 5: Layers of planning

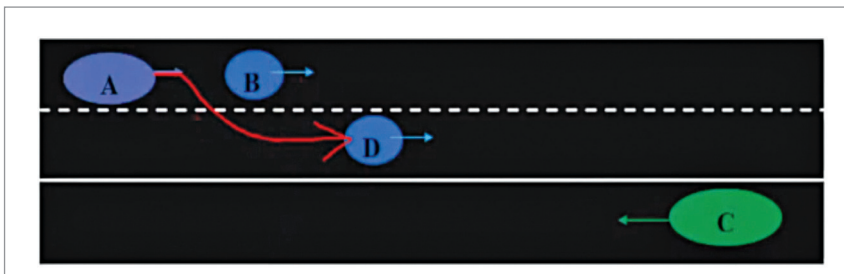


Fig. 6: Vehicles travelling on a road on both lanes of a road

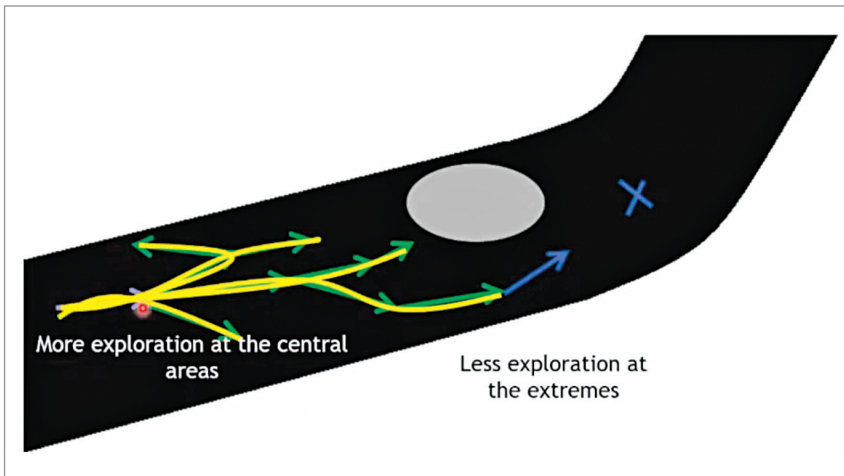


Fig. 7: Rapidly-exploring random tree

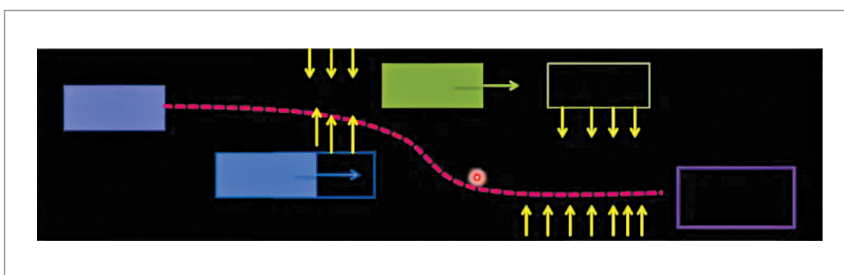


Fig. 8: Self-simulation in case of dynamic obstacles

like as shown by the red arrow line. With an adapted version of a pre-computed lane changing trajectory, an intelligent software controls the vehicle depending on the changes with respect to others.

When a general obstacle is present in front, then humans can calculate the distance pretty fast. A self-driving car does the same by inferring all possible options and finally selecting the most suitable path (shown in blue in Fig. 7). The algorithm is called rapidly-exploring random tree, which generates all possible options represented as a tree.

In case of dynamic obstacles, that is, vehicles that can move (Fig. 8), let us say that trajectory has been built. Unfortunately, the surrounding cars can also move, making the trajectory infeasible.

To counter the uncertainties, the vehicle does a self-simulation with the help of forward simulation technology of various possibilities. In the process, (a simulated future virtual avatar of) each vehicle tries to avoid/repel (a simulated future avatar of) the other vehicle from coming near it. This leads to finally selecting a path, which has the least amount of avoidance/repulsion.

Currently, self-driving cars will never overtake. These have been designed just to simply follow other vehicles on the road, avoid static obstacles, and change lanes, if necessary.

In conclusion, a self-driving car or autonomous vehicle is something that can understand what is going around, use that understanding to determine its current position, map whatever it sees around both at the global and local level, do strategic decision making (overtake, change lane, or avoid vehicles), and finally operate the braking and throttle mechanisms.

There are a lot of problems yet to be solved. Only when everybody concerned comes together on a common platform, can pressing issues be worked out. **EFY**

This article is based on the talk 'Autonomous Vehicles: Tech Making It Possible' by Dr Rahul Kala, assistant professor, Centre Of Intelligent Robotics, IIT-Allahabad during February edition of Tech World Congress and India Electronics Week 2021